



Cambridge International AS & A Level

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MATHEMATICS**9709/32**

Paper 3 Pure Mathematics 3

October/November 2025**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

1 (a) Sketch the graph of $y = |x + 3a|$, where a is a positive constant.

[1]

(b) Hence or otherwise solve the inequality $|x + 3a| > a - 2x$.

[2]

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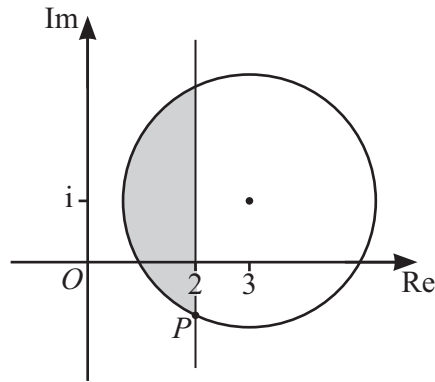




- 2** Solve the equation $3 \times 2^{x+1} = 4 \times 3^{2x-3}$. Give your answer correct to 3 significant figures. [4]

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3



The shaded region in the Argand diagram, bounded by a line and a circle, represents the complex numbers z satisfying

$$\operatorname{Re} z \leq 2 \text{ and } |z - (3 + i)| \leq 2.$$

The point P shown on the diagram is one of the points of intersection of the line and the circle.

- (a) Find the complex number represented by the point P . Give your answer in the form $x + iy$, where x and y are real and exact. [2]

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- (b) Find the greatest value of $\arg z$ for points in the shaded region. [3]

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[6]

[illegible]

- 5 (a) It is given that $f(x) = (x - a)^2 g(x)$, where $f(x)$ and $g(x)$ are polynomials.

Show that $(x - a)$ is a factor of $f'(x)$. [2]

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- (b) It is given that $(x - 3)^2$ is a factor of $2x^3 - 4x^2 + px + q$, where p and q are constants.

Find the values of p and q . [5]

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- 6 (a) By sketching a suitable pair of graphs, show that the equation $\cot 2x = 2 \sin 2x - 1$ has exactly one root in the interval $0 < x < \frac{1}{2}\pi$. [2]

- (b) Show by calculation that the root is in the interval $0.4 < x < 0.6$. [2]

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- (c) Use the iterative formula $x_{n+1} = \frac{1}{2} \tan^{-1} \left(\frac{1}{2 \sin 2x_n - 1} \right)$ to calculate the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

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- (b) Hence find the coordinates of the points on the curve at which the normal is parallel to the y -axis. [4]

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(b) Hence find the exact value of $\int_0^{\frac{1}{4}\pi} \cos^3 x \sin 4x \, dx$.

[5]

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- (b)** Hence find the exact value of $\int_{-a}^a f(x) dx$. Give your answer in the form $a(p + \ln q)$, where p and q are rational. [4]

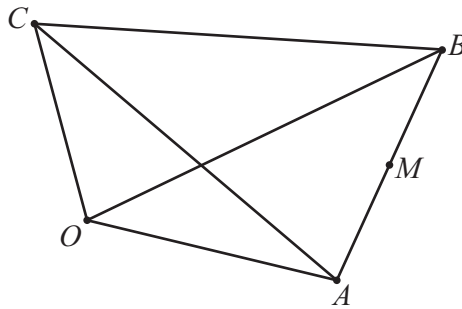
[4]





- (b) Given that $h = 0$ when $t = 0$, find the time taken for the depth of the water in the tank to reach 20 cm. [5]

[illegible]



Relative to the origin O , the position vectors of the points A , B and C are

$$\overrightarrow{OA} = 4\mathbf{i} - 2\mathbf{j}, \overrightarrow{OB} = 2\mathbf{i} + 8\mathbf{j} + 4\mathbf{k}, \text{ and } \overrightarrow{OC} = -2\mathbf{i} + 6\mathbf{k}.$$

The midpoint of AB is M , as shown in the diagram.

- (a) Find the vectors $\overrightarrow{MB}$ and $\overrightarrow{MC}$. [2]

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- (b) Calculate the exact value of the cosine of angle CMB . [3]

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Additional page

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[illegible]

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